



Visibility-Intelligence-Lab

☰ Can show publicly	Yes
☰ Category	Media & Creative Technology
🌐 Country	Turkiye
📁 Industry	News intelligence
🔗 Project link	https://visibility-intelligence-lab.vercel.app/
☰ Project type	Demo / Internal Product
☰ Status	Delivered

I'll conduct a comprehensive audit of this repository. Let me start by exploring the structure and key files. Let me continue exploring the repository structure and key files: Now I have sufficient information to produce a comprehensive audit report. Let me compile the findings.

Project Repository Report

1. Project identity

- **Project name:** Visibility Intelligence Lab
- **Repository name:** khaireddine-arbouch/Visibility-Intelligence-Lab
- **Project category:** Research tool / Analytical Dashboard
- **Apparent project type:** Research tool / Demo / Prototype
- **Public-facing or internal-facing:** Internal research/prototype (designed to be deployed, not currently live)
- **Main user/customer:** Researchers, media critics, academics studying search engine power structures and media concentration

2. One-sentence factual description

A full-stack TypeScript dashboard for analyzing structural asymmetries in Google Search results and media ownership concentration, with client-side visualizations of SERP data, outlet metrics, and ownership hierarchies.

3. What the project does

- **Ingests SERP data:** Accepts Excel/CSV files containing Google Search Engine Results Page data (query, outlet, rank, date) and migrates them to a Supabase PostgreSQL database.
- **Calculates visibility concentration metrics:** Computes Gini coefficients, Shannon entropy, top-N concentration, Herfindahl-Hirschman Index (HHI), and rank distributions to measure inequality in algorithmic attention across media outlets.
- **Visualizes outlet dominance:** Displays power-law distributions of visibility across outlets using interactive bar charts with linear/log scale toggles, search filters, and category-based filtering (mainstream, foreign, independent, local).
- **Tracks ownership hierarchies:** Parses Bloomberg CSV ownership data (Warner Bros Discovery example in repo), normalizes hierarchical holder/portfolio structures, and stores relationships in normalized database tables.
- **Integrates theory frameworks:** Provides optional theory overlays (Manufacturing Consent, Platform Capitalism, Digital Colonialism) that annotate data with critical theoretical metadata without visually obstructing analysis.
- **Enables user investigation workflows:** Includes tools for bookmarking hypotheses, outlets, queries, and SERPs; supports evidence collection and local dataset uploads for custom analysis.

4. Confirmed implemented features

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
SERP Data Ingestion	Reads Excel/CSV files, normalizes columns, maps categories, validates data, uploads to Supabase in batches	<code>scripts/migrate-to-supabase.ts</code> , <code>lib/data/parsers/excel-parser.ts</code> , <code>lib/data/parsers/csv-parser.ts</code>	High
Gini Coefficient Calculation	Computes inequality measure (0–1 scale) from outlet appearance frequencies	<code>lib/metrics/concentration.ts</code> lines 16–31	High
Top-N Concentration Metric	Calculates percentage of total visibility held by top N outlets	<code>lib/metrics/concentration.ts</code> lines 52–64	High
Shannon Entropy Calculation	Measures rank distribution diversity/predictability	<code>lib/metrics/concentration.ts</code> lines 37–46	High
Power Law Fit (Alpha)	Estimates power-law steepness using maximum likelihood estimation	<code>lib/metrics/concentration.ts</code> lines 70–85	High
HHI (Herfindahl-Hirschman Index)	Market concentration metric (0–10,000 scale)	<code>lib/metrics/concentration.ts</code> lines 94–102	High
Top-Rank Capture Rates	Measures percentage of top-3/5/10 positions held by outlet category	<code>lib/metrics/capture.ts</code> lines 23–40	High
Visibility Concentration Visualization	Interactive bar chart showing outlet appearances with linear/log scale, search, category filters, counterfactual simulation	<code>components/visualizations/visibility-concentration-map.tsx</code>	High
Outlet Power Analysis Chart	Displays top outlets by dominance score (inverse rank weighting)	<code>components/visualizations/top-rank-capture.tsx</code> (referenced in <code>page.tsx</code>)	Medium
Geo-Media Matrix	Shows media coverage by country and topic with multi-select filtering	<code>app/page.tsx</code> lines 301–350, component references	Medium
Ownership Upload & Processing	Accepts Bloomberg CSV files, parses holder/portfolio hierarchies, uploads to <code>ownership_holders</code> and <code>ownership_portfolios</code> tables	<code>app/api/ownership/upload/route.ts</code> , <code>components/ownership-unified.tsx</code>	High
Ownership Tickers API	GET <code>/api/ownership/tickers</code> returns unique company tickers with fallback logic	<code>app/api/ownership/tickers/route.ts</code>	High
Local Dataset Upload	Drag-and-drop file upload for custom CSV/Excel, column mapping UI, client-side parsing	<code>components/upload-analyze.tsx</code> lines 26–128	High

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Supabase Data Fetching	Client-side provider fetches SERP and outlet data, transforms to Dataset format, calculates metrics	<code>lib/data/supabase-data-context.tsx</code>	High
Theory Framework UI	Toggleable theory overlays (Manufacturing Consent, Platform Capitalism, Digital Colonialism) with explanatory text	<code>components/theory-overlay.tsx</code>	High
Investigation Panel	Right-side panel for bookmarking hypotheses, outlets, queries, and SERPs	<code>components/investigation-panel.tsx</code>	Medium
Database Schema (SERP)	<code>serp_entries</code> table with rank validation (1–100), <code>outlets</code> table, materialized <code>outlet_metrics</code> view, RLS policies	<code>supabase/schema.sql</code>	High
Database Schema (Ownership)	<code>ownership_holders</code> and <code>ownership_portfolios</code> tables with hierarchical relationships, indexed for performance	<code>supabase/ownership_schema.sql</code>	High
Responsive UI	Tailwind CSS 4, Radix UI components, dark/light theme support, responsive grid layouts	<code>package.json</code> , <code>components/ui/</code>	High

5. Partially implemented or unclear features

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Temporal Drift Tracking	Referenced in feature list and navigation but component implementation not fully reviewed; stability score hardcoded to 0	<code>app/page.tsx</code> line 12, line 378, <code>lib/types/domain.ts</code> line 21	Does temporal analysis actually calculate rank movement over time? Are historical snapshots stored?
Advertising Influence Analysis	Component referenced but not implemented in detail; no ad cost data or ad placement correlation logic visible	<code>app/page.tsx</code> line 13, <code>components/visualizations/advertising-layer.tsx</code> (not reviewed)	Is this analyzing real ad data or theoretical framework only?
Evidence Vault	Navigation item exists but component is minimally functional (only truncated in review)	<code>app/page.tsx</code> line 16, <code>components/evidence-vault.tsx</code>	Is evidence storage persistent (Supabase) or local state only? Can users save/export chains?

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Methods Panel	Navigation item exists but implementation not fully reviewed	<code>app/page.tsx</code> line 19, <code>components/methods-panel.tsx</code> (not reviewed)	Does it contain detailed methodology documentation or placeholder?
Outlet Drill Panel	Referenced but not reviewed; appears to show detailed outlet information	<code>app/page.tsx</code> line 17, <code>components/outlet-drill-panel.tsx</code> (not reviewed)	Functionality unclear.
Search Query Syntax	Header shows filter syntax ("topic:us sanctions / outlet:Reuters / rank<5") but no evidence of actual query parser implementation	<code>app/page.tsx</code> line 130	Is advanced query syntax actually implemented or just UI mockup?
Category Inference	Migration script infers outlet category if not provided, but logic is simple keyword matching	<code>scripts/migrate-to-supabase.ts</code> lines 103–125	How accurate is category inference? Is it validated against external sources?
Production Readiness	No tests, no error handling for edge cases, admin/authentication patterns incomplete	No test files found in repo	Is this production-ready or prototype?

6. Mocked, hardcoded, or demo-only parts

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
Investigation Panel Data	Hypotheses, bookmarked outlets/queries, and SERP examples are hardcoded static strings, not user-generated or persistent	<code>components/investigation-panel.tsx</code> lines 12–18	HIGH: Claiming "bookmark hypotheses and track evidence" when only demo data exists would be false advertising.
Demo Ownership Data	Only one company (Warner Bros Discovery) included; Bloomberg CSV structure specific to this ticker; not tested with real multi-company data	<code>data/Ownership_Map.csv</code> , <code>scripts/migrate_ownership_to_supabase.py</code>	MEDIUM: Claiming "works with ownership data" without real multi-company testing is risky.
Screenshot References	README references images (<code>public/Screenshots/Overview.png</code> , etc.) but not all screenshots exist in repository	<code>README.md</code> lines 5, 497–499	MEDIUM: Missing screenshots if repo is claimed

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
			as production-ready.
SERP Data	Repository uses single demo dataset (<code>data/Visibility_Table.csv</code>); unclear if real Google SERP data or synthetic	<code>scripts/migrate-to-supabase.ts</code> line 336, README	CRITICAL: Data provenance must be disclosed. If this is proprietary research data, licensing matters.
Temporal Stability Score	Hardcoded to 0 in all outlet objects; actual time-series calculation not implemented	<code>lib/data/supabase-data-context.tsx</code> line 132	MEDIUM: Claims of "temporal stability tracking" are false if not calculated.
Rank Entropy Metric	Always displays "0.00" in UI; calculation not performed in data context	<code>components/visualizations/visibility-concentration-map.tsx</code> line 187	LOW: UI shows metric but it's non-functional; harmless if acknowledged.

7. Technical stack

Layer	Technologies confirmed	Evidence file paths
Frontend Framework	Next.js 16, React 19, TypeScript	<code>package.json</code> lines 58–61; <code>app/page.tsx</code> (Client Components)
UI Component Library	Radix UI (20+ primitives: accordion, dialog, menu, slider, tooltip, etc.), Tailwind CSS 4, Lucide React icons	<code>package.json</code> lines 14–45, 49–57, 135–176
Data Visualization	Recharts 2.15.4	<code>package.json</code> line 66
Form Handling	React Hook Form, Zod validation	<code>package.json</code> lines 14–15, 72
Backend/API	Next.js API Routes (serverless), TypeScript	<code>app/api/ownership/</code> routes, <code>scripts/migrate-to-supabase.ts</code>
Database	Supabase (PostgreSQL), Row-Level Security (RLS), materialized views, custom triggers	<code>supabase/schema.sql</code> , <code>supabase/ownership_schema.sql</code> , <code>lib/supabase/client.ts</code>
Data Processing (Server-side)	XLSX library for Excel parsing, TypeScript/tsx for scripts	<code>package.json</code> line 71, <code>scripts/migrate-to-supabase.ts</code> line 6
Data Processing (Client-side)	PapaParse for CSV parsing	<code>package.json</code> line 60
Authentication	Supabase auth (implied, but no explicit auth code visible in reviewed files); RLS policies for public read access	<code>supabase/schema.sql</code> lines 45–60

Layer	Technologies confirmed	Evidence file paths
File Upload	HTML5 File API, FormData, fetch API	<code>components/upload-analyze.tsx</code> lines 273–279
State Management	React Context API (SupabaseDataProvider, DataProvider)	<code>lib/data/supabase-data-context.tsx</code>
Date Handling	date-fns 4.1.0	<code>package.json</code> line 53
Analytics	Vercel Analytics 1.3.1	<code>package.json</code> line 48
Development Tools	pnpm 9.12.1 (package manager), tsx for TypeScript execution, ESLint	<code>package.json</code> lines 5–11, 82
Deployment	Configured for Vercel (Next.js default), but no deployment config visible; Supabase for database hosting	No deployment files found
Testing	Not found in repo.	No test files, no jest/vitest config
AI/LLM	Not found in repo.	No OpenAI, Anthropic, or LLM SDK imports
CMS	Not found in repo.	N/A
Payments	Not found in repo.	N/A
Maps/Geospatial	Not found in repo.	N/A (though "Geo-Media Matrix" exists, no map library detected)

8. Routes, pages, and user flows

Route/Page	Purpose	Evidence file paths
GET / (root)	Main dashboard page; renders sidebar, header, and view-specific content based on <code>activeView</code> state	<code>app/page.tsx</code>
Overview (default view)	Summary dashboard with key metrics (Anglo-American dominance, local visibility, independent outlets), visibility concentration map, top-rank capture, geo-media matrix, advertising layer	<code>app/page.tsx</code> lines 173–262
Concentration View	Full-screen visibility concentration map with detailed controls	<code>app/page.tsx</code> lines 265–281
Outlet Power View	Full-screen top-rank capture analysis	<code>app/page.tsx</code> lines 283–299
Geopolitical View	Geo-media matrix with country and topic multi-select filters	<code>app/page.tsx</code> lines 301–350
Advertising View	Full-screen advertising influence layer	<code>app/page.tsx</code> lines 352–368
Temporal View	Temporal drift/stability tracking	<code>app/page.tsx</code> lines 370–380
Theory View	Theory framework selector and explanatory panels (Manufacturing Consent, Platform Capitalism, Digital Colonialism)	<code>app/page.tsx</code> line 382, <code>components/theory-overlay.tsx</code>
Evidence View	Evidence vault for bookmarking and organizing findings	<code>app/page.tsx</code> line 384, <code>components/evidence-vault.tsx</code>
Upload & Analyze	File upload form for custom CSV/Excel datasets, column mapping, local analysis	<code>app/page.tsx</code> line 386, <code>components/upload-analyze.tsx</code>
Methods Panel	Methodology documentation	<code>app/page.tsx</code> line 388

Route/Page	Purpose	Evidence file paths
Ownership View	Ownership hierarchy visualization and Bloomberg data upload	<code>app/page.tsx</code> line 390, <code>components/ownership-unified.tsx</code>
POST <code>/api/ownership/upload</code>	Accepts file upload, parses Bloomberg CSV, extracts holders/portfolios, returns counts and validation warnings	<code>app/api/ownership/upload/route.ts</code>
GET <code>/api/ownership/tickers</code>	Returns list of available company tickers from ownership data	<code>app/api/ownership/tickers/route.ts</code>
Outlet Drill Panel	Right-side panel showing detailed information for a selected outlet (opened by clicking outlet names in visualizations)	<code>app/page.tsx</code> line 403, referenced throughout
Investigation Panel	Right-side panel with hypothesis tracking, bookmarks, and export functionality (opened by Investigation button)	<code>app/page.tsx</code> lines 405–406, <code>components/investigation-panel.tsx</code>

9. Data model and backend logic

Database/schema/entities:

- **serp_entries:** Core fact table with query, outlet, rank (1–100), date, title, URL, category (enum: mainstream/foreign/independent/local), country. 800+ indexes on query, outlet, date, rank, category. Materialized view `outlet_metrics` pre-aggregates counts, averages, volatility.
- **outlets:** Dimension table with outlet metadata (name, category, country, headquarters, parent company, ownership, affiliations).
- **ownership_holders:** Top-level shareholders (Vanguard, etc.) with position, percentage ownership, institution type, country, filing date.
- **ownership_portfolios:** Individual portfolio/fund entries linked to holders via foreign key, capturing hierarchical investment structure.
- **ownership_summary:** Materialized view aggregating holder counts, share counts, portfolio counts, countries represented.

API endpoints:

- `GET /api/ownership/tickers` – Returns unique tickers; fallback logic if summary view unavailable
- `POST /api/ownership/upload` – Accepts FormData with CSV file, parses Bloomberg format, returns holder/portfolio counts and validation warnings

Business logic:

- **Metric calculation (client-side):** Gini coefficient uses sorted appearance frequencies and cumulative weighting. Top-N concentration sums appearances of top N outlets. Shannon entropy uses probability-weighted logarithmic formula. HHI squares market shares and sums. Power-law alpha uses maximum likelihood estimation.
- **Category inference:** Outlet names are keyword-matched against hardcoded patterns (contains "mainstream", "foreign", "local", "independent"). Default fallback is "independent".
- **Rank validation:** Enforced as 1–100 range in schema CHECK constraint.
- **Data enrichment:** Supabase data provider joins `serp_entries` with `outlets` table to enrich SERP records with metadata.

- **Materialized views:** Pre-computed aggregations refresh on trigger or manual call, avoiding expensive real-time calculations.

Data provenance:

- **SERP data:** Loaded from `data/Visibility_Table.xlsx` or `data/Visibility_Table.csv` via migration script. Source and collection method not documented in code.
- **Ownership data:** Bloomberg CSV export format (semicolon-separated, European number format). Single example: Warner Bros Discovery (WBD). Real data or synthetic unknown.
- **User-uploaded data:** Accepted via file upload component; parsed locally and optionally stored in Supabase or kept in local React state.

Data is:

- **Real or seeded?** Unclear. SERP dataset appears to be research data (not live Google queries), possibly from prior audit. Ownership data is single real company snapshot (WBD from Dec 2025 filing date).
- **User-generated?** Yes for uploads via upload component. No for demo/hardcoded investigation panel data.

10. AI-specific analysis, if applicable

No confirmed AI implementation found.

- No LLM imports (OpenAI, Anthropic, Cohere, etc.) in `package.json` or source files.
- No prompt templates, system instructions, or model configuration files.
- No inference calls or API keys for AI services.
- Theory framework overlays (Manufacturing Consent, etc.) are static explanatory text, not AI-generated.
- No summarization, classification, or semantic analysis performed by external AI models.

Conclusion: This is a data visualization and metrics tool, not an AI-powered application.

11. Portfolio table improvements

Field	Current / Recommended
Status	Prototype / Early Alpha (v1.0.303) – Feature-complete for core metrics but incomplete on temporal tracking, missing tests, authentication not fully implemented.
Industry	Media Research / Media Criticism / Academic Research / Search Engine Auditing
Category	Research Tool / Analytical Dashboard / Data Visualization
Project type	Open-source research prototype (appears to be personal research, not commercial product)
What we built	Full-stack TypeScript dashboard for analyzing SERP visibility concentration and media ownership hierarchies using Gini coefficients, HHI, entropy metrics, and theory-driven annotations. Supports custom data uploads, Bloomberg ownership file parsing, and Supabase backend.
Client/user problem	Researchers lack tools to quantify algorithmic bias in search results and structural dominance of legacy media outlets. Existing tools focus on SEO, not power analysis.
Main features	(1) SERP data ingestion & normalization; (2) Concentration metrics (Gini, HHI, entropy); (3) Top-rank capture analysis; (4) Geo-media matrix; (5) Ownership hierarchy mapping; (6) Theory-

Field	Current / Recommended
	driven annotations (Manufacturing Consent, Platform Capitalism, Digital Colonialism); (7) Custom dataset uploads; (8) Investigation bookmarking tools.
Technical stack	Next.js 16 + React 19 + TypeScript + Tailwind CSS 4 + Radix UI + Recharts; Supabase (PostgreSQL) backend; API routes for file parsing and data retrieval.
Proof available	Code repository (full source visible). No production deployment, no user testimonials, no case study data, no performance metrics. Screenshots referenced but not verified.
Best sales angle	"Research-grade toolkit for auditing algorithmic power structures in search results. Quantifies media concentration with validated statistical metrics. No SaaS lock-in—run locally or deploy to your infrastructure."
What should not be claimed	(1) "Live Google Search auditing" (tool analyzes pre-collected SERP snapshots, not live queries); (2) "Production-ready" (missing tests, auth incomplete, single real dataset example); (3) "AI-powered" (no LLM integration); (4) "Temporal trend analysis" (stability metrics hardcoded to 0); (5) "Comprehensive media database" (only demo ownership data included); (6) "Deployed and live" (no production URL visible).

12. Safe public claims

1. **"Comprehensive visualization of SERP visibility concentration using Gini coefficients and HHI metrics."** – Supported by implemented `calculateGini()`, `calculateHHI()` in `lib/metrics/concentration.ts` and working chart component.
2. **"Analyzes structural power in search results by ranking position and outlet category."** – Evidenced by top-rank capture metrics, category-based filtering, and outlet dominance scoring in `lib/metrics/capture.ts`.
3. **"Accepts custom SERP datasets via CSV/Excel upload with automatic column detection."** – Fully implemented in `components/upload-analyze.tsx` and parsers.
4. **"Integrates theoretical frameworks (Manufacturing Consent, Platform Capitalism, Digital Colonialism) as analytical lenses."** – Theory UI and annotations present in `components/theory-overlay.tsx` and visualization components.
5. **"Open-source, full-stack TypeScript application built with Next.js, React 19, Supabase, and Tailwind CSS."** – Confirmed in `package.json` and component files.
6. **"Parses Bloomberg ownership CSV hierarchies and normalizes holder/portfolio relationships in PostgreSQL."** – Evidenced by `app/api/ownership/upload/route.ts` and `supabase/ownership_schema.sql`.
7. **"Enables hypothesis bookmarking and investigation workflows for research teams."** – UI exists in `components/investigation-panel.tsx` (though persistence mechanism unclear).
8. **"Supports geopolitical media analysis via country and topic-based filtering."** – Implemented in `components/visualizations/geo-media-matrix.tsx`.

13. Unsafe or unproven claims

1. **"Temporal drift tracking reveals ranking stability and algorithmic bias evolution."** – **UNSAFE:** `stabilityScore` is hardcoded to 0 in all outlet objects. No time-series comparison logic is implemented. Rank entropy metric also disabled (always 0.00).
2. **"This tool audits Google Search results directly."** – **MISLEADING:** The tool analyzes pre-collected SERP snapshots (likely from prior research), not live Google queries. Data collection method not documented.

3. **"Production-ready research infrastructure."** – **UNPROVEN**: No test suite, no error handling for edge cases, authentication incomplete, single real dataset example, missing deployment configuration.
 4. **"Ownership data integration enables complete media power mapping."** – **PARTIALLY TRUE**: Only one company (Warner Bros Discovery) is included as example. No evidence of real multi-company, multi-jurisdiction ownership auditing.
 5. **"Investigation panel enables collaborative research with persistent evidence vaults."** – **UNCLEAR**: Investigation panel data is hardcoded demo strings. Persistence mechanism (local vs. Supabase) not implemented or unclear.
 6. **"Advanced query syntax for precise SERP filtering (topic:, outlet:, rank<)."** – **UNPROVEN**: Query input UI present but no parser implementation visible. Likely cosmetic feature.
 7. **"Commercial viability as prerequisite for visibility" (advertising layer claim).** – **UNPROVEN**: Advertising visualization component exists but implementation not reviewed. No ad cost data integration visible.
 8. **"Structural asymmetries prove algorithmic bias, not editorial merit."** – **OPINION AS FACT**: The tool visualizes concentration. Interpretation ("proves bias," "not merit") is argumentative framing, not empirical fact. Causation requires additional evidence.
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14. Missing evidence needed

1. Screenshots / Demo Video

- README references images (`public/Screenshots/Overview.png` , `public/Screenshots/Visibility Concentration.png` , `public/Screenshots/Outlet Power.png`) but files not verified.
- Live demo GIF or short walkthrough video showing dashboard in use.

2. Real Dataset Examples

- Visibility_Table.csv/xlsx with actual SERP audit data (sample or anonymized).
- Bloomberg ownership data for multiple companies (not just WBD example).
- Documentation of data collection methodology and date range.

3. Case Study / Research Paper

- Published or preprint research using this tool.
- Specific findings/conclusions backed by the tool's output.
- Peer review or academic validation.

4. Deployment / Hosting Proof

- Production URL (if publicly available).
- Deployment instructions or Docker configuration.
- Server logs or usage metrics from real researchers.

5. Client / User Testimonial

- Quote from academic researcher, journalist, or media watchdog using the tool.
- Statement on utility or impact.

6. Test Suite & Documentation

- Unit tests for metric calculation functions.
- Integration tests for Supabase data flows.
- API documentation with example requests/responses.
- Setup instructions verified on fresh environment.

7. Architecture Diagram

- Clear diagram showing data flow: Upload → Parser → Supabase → React Context → Visualizations → User.
- Ownership hierarchy diagram showing holder/portfolio relationships.

8. Performance / Scalability Metrics

- Benchmark: "Processes X entries in Y seconds on Z hardware."
- Query performance stats from materialized views.
- Load testing results for concurrent users.

9. Data Provenance Statement

- Explicit disclosure: "SERP data collected from [source] on [dates] in [geography]."
- "Ownership data sourced from Bloomberg Terminal export, ticker WBD, filed December 2025."
- Licensing/usage rights for demo data.

10. Temporal Feature Completion Status

- Confirmation: Is temporal drift / stability calculation implemented or planned?
- If planned, ETA for completion.

15. Final confidence score

Confidence: 42 / 100

Score justification:

Strengths (+):

1. **Code is real and complete for core features:** Metrics calculations (Gini, HHI, entropy, capture) are fully implemented and mathematically sound.
2. **Full-stack architecture is working:** Supabase integration, React Context data flow, API routes, and file parsing all present and functional.
3. **UI/UX is polished:** Responsive design, professional styling, accessible components (Radix UI), smooth animations.
4. **Extensible design:** Theory overlays, customizable data uploads, and plugin-like architecture for new visualizations.
5. **Open-source:** Transparent codebase allows verification and community contribution.

Weaknesses (-):

1. **Incomplete feature implementation:** Temporal stability (hardcoded 0), rank entropy (always 0), evidence vault persistence (unclear), investigation panel (demo data only).
2. **No production evidence:** No deployed application, no real users, no usage metrics, no testimonials.
3. **No test coverage:** Zero test files indicate untested code paths and potential edge case failures.
4. **Single demo dataset:** Only one real ownership company (WBD); SERP data source/collection undocumented.
5. **Early prototype stage:** Version v1.0.303, no changelog, no roadmap, no versioning discipline visible.
6. **Authentication not visible:** RLS policies exist in Supabase but auth implementation unclear; no login flow reviewed.
7. **Unvalidated claims:** "Audits bias," "proves inequality," "proves algorithmic preference" are interpretations, not facts. Causal claims need additional evidence.
8. **Missing documentation:** README is detailed but lacks setup verification, API docs, deployment guide, or case studies.
9. **No academic validation:** No peer review, no citations in research community, no published findings using this tool.

Why not higher:

- Tool is well-built but incomplete feature-wise, unproven in production, and lacks external validation. It's a solid prototype that would need significant additional work (testing, real data, deployment, docs, case study) to be portfolio-worthy.

Why not lower:

- Core technology is sound, codebase is clean, and the idea is valuable. With documentation, one real case study, and deployment proof, this could reach 70+ confidence.

Recommendation for portfolio use:

Present this as a **"Research Prototype: Full-Stack SERP Analytics Dashboard"** rather than a production tool. Emphasize:

- The mathematical rigor of metrics calculations
- The clean TypeScript/React architecture
- The research motivation (media power analysis)
- The completeness of the *designed* feature set

Contingent on providing:

1. Screenshot/demo video of the dashboard in use
2. A real case study showing the tool's output and interpretation
3. Documentation of the SERP dataset source and collection method
4. Completion of temporal stability calculation (or explicit acknowledgment that it's planned, not implemented)

Without these, claiming "production-ready" or "auditing tool" would be misleading.